Lab Exercise 10- Creating and Managing a ReplicaSet in Kubernetes

Objective:

A ReplicaSet in Kubernetes ensures a specified number of Pod replicas are running at any given time. This exercise will guide you through creating a ReplicaSet to maintain the desired state of your application.

- Understand the syntax and structure of a Kubernetes ReplicaSet definition file (YAML).
- Learn how to create and manage a ReplicaSet to ensure application availability.
- Understand how a ReplicaSet helps in scaling applications and maintaining desired states.

Prerequisites

- Kubernetes Cluster: Have a running Kubernetes cluster (locally using Minikube or kind, or a cloud-based service).
- kubectl: Install and configure kubectl to interact with your Kubernetes cluster.
- Basic Knowledge of YAML: Familiarity with YAML format will be helpful for understanding Kubernetes resource definitions.

Step-by-Step Guide

Step 1: Understanding ReplicaSet

A ReplicaSet ensures a specified number of Pod replicas are running at any given time. If a Pod crashes or is deleted, the ReplicaSet creates a new one to meet the defined number of replicas. This helps maintain application availability and ensures that your application can handle increased load by distributing traffic among multiple Pods.

Step 2: Create a ReplicaSet

We'll define a ReplicaSet to maintain three replicas of a simple Nginx web server Pod. Create a YAML file named nginx-replicaset.yaml with the following content:

```yaml
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: nginx-replicaset
spec:
  replicas: 3
  selector:
    matchLabels:
      app: my-app
  template:
    metadata:
      labels:
        app: my-app
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Explanation:

- apiVersion: Defines the API version (apps/v1) used for the ReplicaSet resource.
- kind: Specifies that this resource is a ReplicaSet.
- metadata: Contains metadata about the ReplicaSet, including name.

    - name: The unique name for the ReplicaSet.
- spec: Provides the specification for the ReplicaSet.
    - replicas: Defines the desired number of Pod replicas.
    - selector: Criteria for selecting Pods managed by this ReplicaSet.
        - matchLabels: Labels that Pods must have to be managed by this ReplicaSet.
    - template: Defines the Pod template used for creating new Pods.
        - metadata: Contains metadata for the Pods, including labels.
            - labels: Labels applied to Pods created by this ReplicaSet.
    - spec: Specification for the Pods.
        - containers: Lists the containers that will run in the Pod.
            - name: The unique name of the container within the Pod.
            - image: The Docker image used for the container.
            - ports: Ports exposed by the container.

Step 3: Apply the YAML to Create the ReplicaSet

Use the kubectl apply command to create the ReplicaSet based on the YAML file.

```
kubectl apply -f nginx-replicaset.yaml
```

Verify the ReplicaSet is running and maintaining the desired number of replicas:

```
kubectl get replicaset
```

This command lists all ReplicaSets in the current namespace.

To check the Pods created by the ReplicaSet:

```
kubectl get pods -l app=nginx
```

This command lists all Pods with the label app=nginx.

Step 4: Managing the ReplicaSet

1. Scaling the ReplicaSet

You can scale the number of replicas managed by the ReplicaSet using the kubectl scale command.

```
kubectl scale --replicas=5 replicaset/nginx-replicaset
```

This command scales the ReplicaSet to maintain 5 replicas. Verify the scaling operation:

```
kubectl get pods -l app=nginx
```

You should see that the number of Pods has increased to 5.

2. Updating the ReplicaSet

If you need to update the Pod template (e.g., to use a different Docker image version), modify the YAML file and apply it again. For instance, change the image to a specific version of Nginx:

```
spec:
  template:
    spec:
      containers:
      - name: nginx
        image: nginx:1.19.3  # Change to a specific version
```

Apply the changes:

```
kubectl apply -f nginx-replicaset.yaml
```

Check the status to ensure the Pods are updated:

```
kubectl get pods -l app=nginx
```

Note: Updating a ReplicaSet doesn't automatically replace existing Pods with new ones. In practice, you often create a new ReplicaSet or Deployment for updates.

3. Deleting the ReplicaSet

To clean up the ReplicaSet and its Pods, use the kubectl delete command:

```
kubectl delete -f nginx-replicaset.yaml
```

This command deletes the ReplicaSet and all the Pods managed by it.

```
C:\Users\NK>(echo apiVersion: apps/v1 & echo kind: ReplicaSet & echo metadata: & echo   name: nginx-replicaset & echo spec: & echo   replicas: 3 & echo   se
lector: & echo     matchLabels: & echo       app: my-app & echo   template: & echo     metadata: & echo       labels: & echo         app: my-app & echo
spec: & echo       containers: & echo       - name: nginx & echo         image: nginx:latest & echo         ports: & echo         - containerPort: 80) > ngi
nx-replicaset.yaml

C:\Users\NK>type nginx-replicaset.yaml
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: nginx-replicaset
spec:
  replicas: 3
  selector:
    matchLabels:
      app: my-app
  template:
    metadata:
      labels:
        app: my-app
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80

C:\Users\NK>minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured


C:\Users\NK>kubectl apply -f nginx-replicaset.yaml
replicaset.apps/nginx-replicaset created
```

```
nginx-replicaset   3         3         1       4s

C:\Users\NK>kubectl get pods -l app=my-app
NAME                    READY   STATUS              RESTARTS   AGE
nginx-replicaset-8c6zm   1/1     Running             0          7s
nginx-replicaset-8h6fw   1/1     Running             0          7s
nginx-replicaset-mb6ss   0/1     ContainerCreating   0          7s

C:\Users\NK>kubectl scale --replicas=5 replicaset/nginx-replicaset
replicaset.apps/nginx-replicaset scaled

C:\Users\NK>kubectl get pods -l app=my-app
NAME                    READY   STATUS              RESTARTS   AGE
nginx-replicaset-8c6zm   1/1     Running             0          14s
nginx-replicaset-8h6fw   1/1     Running             0          14s
nginx-replicaset-fjgsm   0/1     ContainerCreating   0          3s
nginx-replicaset-kkqkm   0/1     ContainerCreating   0          3s
nginx-replicaset-mb6ss   1/1     Running             0          14s

C:\Users\NK>kubectl scale --replicas=2 replicaset/nginx-replicaset
replicaset.apps/nginx-replicaset scaled

C:\Users\NK>kubectl get pods -l app=my-app
NAME                    READY   STATUS    RESTARTS   AGE
nginx-replicaset-8c6zm   1/1     Running   0          24s
nginx-replicaset-8h6fw   1/1     Running   0          24s

C:\Users\NK>kubectl describe replicaset nginx-replicaset
Name:         nginx-replicaset
Namespace:    default
Selector:     app=my-app
Labels:       <none>
Annotations:  <none>
Replicas:     2 current / 2 desired
Pods Status:  2 Running / 0 Waiting / 0 Succeeded / 0 Failed
Pod Template:
  Labels:  app=my-app
  Containers:
   nginx:
    Image:         nginx:latest
    Port:          80/TCP
```

```
nginx-replicaset-8c6zm   1/1     Running   0          24s
nginx-replicaset-8h6fw   1/1     Running   0          24s

C:\Users\NK>kubectl describe replicaset nginx-replicaset
Name:         nginx-replicaset
Namespace:    default
Selector:     app=my-app
Labels:       <none>
Annotations:  <none>
Replicas:     2 current / 2 desired
Pods Status:  2 Running / 0 Waiting / 0 Succeeded / 0 Failed
Pod Template:
  Labels:  app=my-app
  Containers:
   nginx:
    Image:         nginx:latest
    Port:          80/TCP
    Host Port:     0/TCP
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Events:
  Type    Reason            Age   From                   Message
  ----    ------            ----  ----                   -------
  Normal  SuccessfulCreate  27s   replicaset-controller  Created pod: nginx-replicaset-8c6zm
  Normal  SuccessfulCreate  27s   replicaset-controller  Created pod: nginx-replicaset-8h6fw
  Normal  SuccessfulCreate  27s   replicaset-controller  Created pod: nginx-replicaset-mb6ss
  Normal  SuccessfulCreate  16s   replicaset-controller  Created pod: nginx-replicaset-fjgsm
  Normal  SuccessfulCreate  16s   replicaset-controller  Created pod: nginx-replicaset-kkqkm
  Normal  SuccessfulDelete  9s    replicaset-controller  Deleted pod: nginx-replicaset-mb6ss
  Normal  SuccessfulDelete  9s    replicaset-controller  Deleted pod: nginx-replicaset-kkqkm
  Normal  SuccessfulDelete  9s    replicaset-controller  Deleted pod: nginx-replicaset-fjgsm

C:\Users\NK>kubectl delete -f nginx-replicaset.yaml
replicaset.apps "nginx-replicaset" deleted from default namespace

C:\Users\NK>kubectl get replicaset
No resources found in default namespace.
```